

POPULATION PARAMETERS AND DISTRIBUTION MOMENTS

PROBABILITY AND STATISTICS FOUNDATIONS — PART 2

Instructor: Tudor Schlanger

Master's in Asset Management
Yale School of Management

OBJECTIVES FOR THIS SESSION

1. Expose you to ideas and terminology
2. Teach you how we map the real world to math objects
3. Teach you how to interpret math objects using real world examples

OUTLINE

1. Random variables and distributions

How do we describe an uncertain outcome mathematically?

2. **Population parameters and moments**

If we know the distribution, which summary numbers matter?

3. Statistical inference

If we *don't* know the distribution, how do we learn it from data?

QUESTION OF THE LESSON

Assume you have a portfolio comprised of two assets: **stocks** and **bonds**.

They both have uncertain returns, R_s and R_b , with known joint distributions.

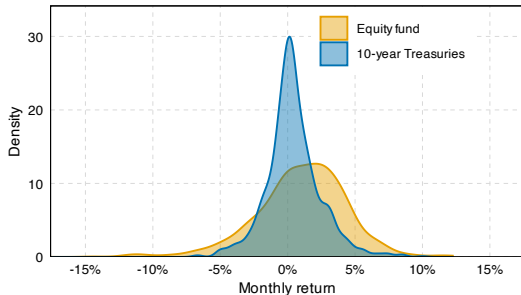
Your allocation is already set: a fraction ω in stocks, the rest in bonds.

QUESTION OF THE LESSON

Assume you have a portfolio comprised of two assets: **stocks** and **bonds**.

They both have uncertain returns, R_s and R_b , with known joint distributions.

Your allocation is already set: a fraction ω in stocks, the rest in bonds.



Monthly returns, 1941–2023. S&P Composite total return from Shiller; 10-year Treasury returns from the course data.

Question

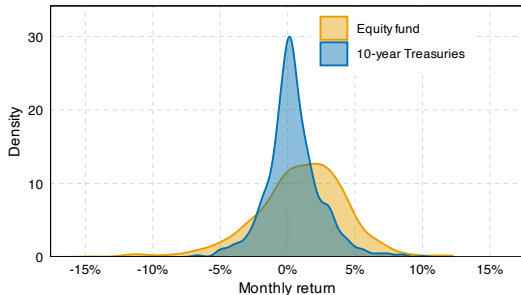
What is the **risk–return profile** of my portfolio?

QUESTION OF THE LESSON

Assume you have a portfolio comprised of two assets: **stocks** and **bonds**.

They both have uncertain returns, R_s and R_b , with known joint distributions.

Your allocation is already set: a fraction ω in stocks, the rest in bonds.



Monthly returns, 1941–2023. S&P Composite total return from Shiller; 10-year Treasury returns from the course data.

Question

What is the **risk–return profile** of my portfolio?

We can't read that off the distributions! We need an objective.

PREFACING THE ANSWER

What matters is the portfolio return, R_p , which is itself a random variable:

$$R_p = \omega R_s + (1 - \omega) R_b$$

where ω is the weight on stocks.

PREFACING THE ANSWER

What matters is the portfolio return, R_p , which is itself a random variable:

$$R_p = \omega R_s + (1 - \omega) R_b$$

where ω is the weight on stocks.

The objective: “Investors like high returns, but dislike high volatility”

PREFACING THE ANSWER

What matters is the portfolio return, R_p , which is itself a random variable:

$$R_p = \omega R_s + (1 - \omega)R_b$$

where ω is the weight on stocks.

The objective: “Investors like high returns, but dislike high volatility”

Translating this into pseudo-math — the **return-to-risk ratio**:

$$\frac{\text{Return}}{\text{Risk}} = \frac{\text{Average return of } R_p}{\text{Volatility of } R_p}.$$

PREFACING THE ANSWER

What matters is the portfolio return, R_p , which is itself a random variable:

$$R_p = \omega R_s + (1 - \omega) R_b$$

where ω is the weight on stocks.

The objective: “Investors like high returns, but dislike high volatility”

Translating this into pseudo-math — the **return-to-risk ratio**:

$$\frac{\text{Return}}{\text{Risk}} = \frac{\text{Average return of } R_p}{\text{Volatility of } R_p}.$$

Translating pseudo-math to math object:

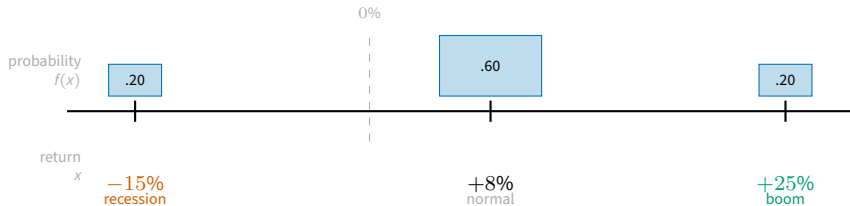
Average return of $R_p \rightarrow$ Expected value

Volatility of $R_p \rightarrow$ Variance

EXPECTED VALUE, SEEN ON A BEAM

An analyst assigns three scenarios to next year's return: **recession**, normal year, **boom**.

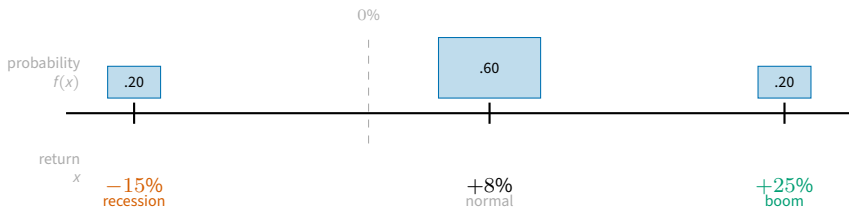
Picture the probabilities as weights on a beam: the expected value is where it **balances**.



EXPECTED VALUE, SEEN ON A BEAM

An analyst assigns three scenarios to next year's return: **recession**, normal year, **boom**.

Picture the probabilities as weights on a beam: the expected value is where it **balances**.

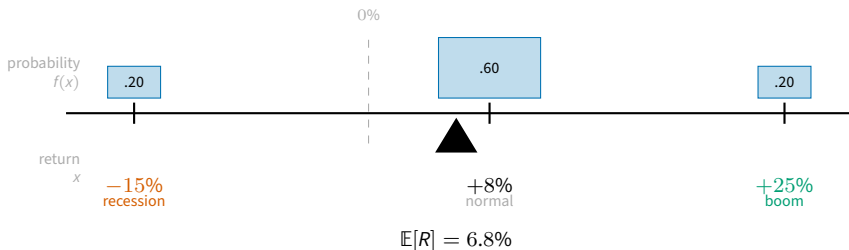


$0.2(-15) + 0.6(8) + 0.2(25) = 6.8$. The balance point sits a little *left* of the “normal” scenario, because the **recession** is further from it than the **boom** is. **Distance matters as much as probability.**

EXPECTED VALUE, SEEN ON A BEAM

An analyst assigns three scenarios to next year's return: **recession**, normal year, **boom**.

Picture the probabilities as weights on a beam: the expected value is where it **balances**.



$0.2(-15) + 0.6(8) + 0.2(25) = 6.8$. The balance point sits a little *left* of the “normal” scenario, because the **recession** is further from it than the **boom** is. **Distance matters as much as probability.**

EXPECTED VALUE, IN MATH

Discrete X

Taking values $x_1, \dots, x_k$:

$$\mathbb{E}[X] = \sum_{j=1}^k x_j f(x_j).$$

Each possible value, **weighted by how likely it is**.

EXPECTED VALUE, IN MATH

Discrete X

Taking values $x_1, \dots, x_k$:

$$\mathbb{E}[X] = \sum_{j=1}^k x_j f(x_j).$$

Each possible value, **weighted by how likely it is**.

Continuous X

The sum becomes an integral:

$$\mathbb{E}[X] = \int x f(x) dx.$$

EXPECTED VALUE, IN MATH

Discrete X

Taking values $x_1, \dots, x_k$:

$$\mathbb{E}[X] = \sum_{j=1}^k x_j f(x_j).$$

Continuous X

The sum becomes an integral:

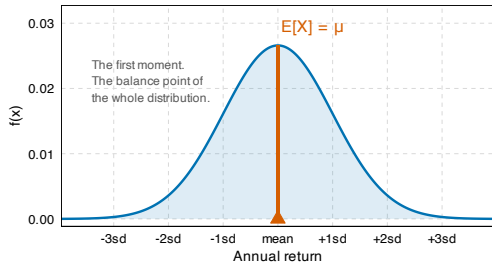
$$\mathbb{E}[X] = \int x f(x) dx.$$

Each possible value, **weighted by how likely it is**.

We usually denote the expected value as a constant $\mu = \mathbb{E}[X]$.

The expected value is also referred to as the “first moment.”

EXPECTED VALUE, SEEN ON A DISTRIBUTION



Continuous X

The sum becomes an integral:

$$\mathbb{E}[X] = \int x f(x) dx.$$

EXPECTED VALUE $\neq$ SAMPLE MEAN

Colloquially, we refer to the expected value as the mean, but let's be more precise.

	Expected value	Sample mean
Symbol	$\mathbb{E}[X]$	$\bar{X}$
Formula	$\sum_j x_j f(x_j)$	$\frac{1}{n} \sum_{i=1}^n x_i$
Built from	the true distribution $f(x)$	n observed data points
Is it random?	No — one fixed number	Yes — a new sample gives a new value

EXPECTED VALUE $\neq$ SAMPLE MEAN

Colloquially, we refer to the expected value as the mean, but let's be more precise.

	Expected value	Sample mean
Symbol	$\mathbb{E}[X]$	$\bar{X}$
Formula	$\sum_j x_j f(x_j)$	$\frac{1}{n} \sum_{i=1}^n x_i$
Built from	the true distribution $f(x)$	n observed data points
Is it random?	No — one fixed number	Yes — a new sample gives a new value

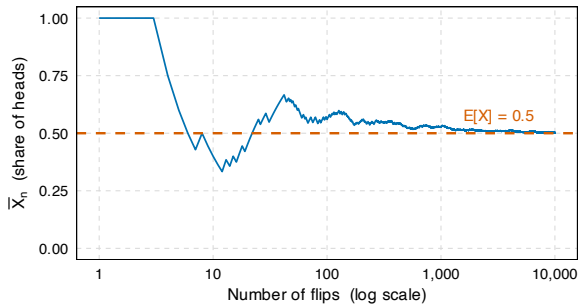
Law of Large Numbers

If $X_1, \dots, X_n$ are **independent** draws from the **same** distribution (i.i.d.), and $\mathbb{E}[X]$ is finite, then

$$\bar{X} \xrightarrow{p} \mathbb{E}[X] \quad \text{as } n \rightarrow \infty.$$

The p over the arrow reads *in probability*: draw enough times and $\bar{X}$ lands within any margin you name of $\mathbb{E}[X]$.

FLIP A COIN UNTIL $\bar{X} \rightarrow \mathbb{E}[X]$



Heads $X = 1$, tails $X = 0$.

$\bar{X}$ is the sample mean of the 0's and 1's. This is the share of heads.

Since coin is fair, $\mathbb{E}[X] = \mathbf{0.5}$.

One run of 10,000 flips of a fair coin, recomputing $\bar{X}$ after every flip. Horizontal axis on a log scale.

LINEARITY OF EXPECTED VALUES

E.1 For any constant c : $\mathbb{E}[c] = c$.

LINEARITY OF EXPECTED VALUES

E.1 For any constant c : $\mathbb{E}[c] = c$.

E.2 For any constants a and b :

$$\mathbb{E}[aX + b] = a \mathbb{E}[X] + b.$$

LINEARITY OF EXPECTED VALUES

E.1 For any constant c : $\mathbb{E}[c] = c$.

E.2 For any constants a and b :

$$\mathbb{E}[aX + b] = a \mathbb{E}[X] + b.$$

E.3 For constants $a_1, \dots, a_n$:

$$\mathbb{E}[a_1X_1 + \dots + a_nX_n] = a_1\mathbb{E}[X_1] + \dots + a_n\mathbb{E}[X_n].$$

LINEARITY OF EXPECTED VALUES

E.1 For any constant c : $\mathbb{E}[c] = c$.

E.2 For any constants a and b :

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b.$$

E.3 For constants $a_1, \dots, a_n$:

$$\mathbb{E}[a_1X_1 + \dots + a_nX_n] = a_1\mathbb{E}[X_1] + \dots + a_n\mathbb{E}[X_n].$$

Question

A fund charges a flat 1% per year. The gross return is R . What's the expected net return?

LINEARITY OF EXPECTED VALUES

E.1 For any constant c : $\mathbb{E}[c] = c$.

E.2 For any constants a and b :

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b.$$

E.3 For constants $a_1, \dots, a_n$:

$$\mathbb{E}[a_1X_1 + \dots + a_nX_n] = a_1\mathbb{E}[X_1] + \dots + a_n\mathbb{E}[X_n].$$

Question

A fund charges a flat 1% per year. The gross return is R . What's the expected net return?

Answer

Using E.2,

$$\mathbb{E}[R - 0.01] = \mathbb{E}[R] - 0.01.$$

Fees come straight off the top of the expected return.

EXPECTED RETURN ON A PORTFOLIO

Question

The portfolio return is

$$R_p = \omega R_s + (1 - \omega)R_b.$$

Consider ω fixed. What is the expected return?

EXPECTED RETURN ON A PORTFOLIO

Question

The portfolio return is

$$R_p = \omega R_s + (1 - \omega) R_b.$$

Consider ω fixed. What is the expected return?

Answer

Using E.3,

$$\mathbb{E}[R_p] = \omega \mathbb{E}[R_s] + (1 - \omega) \mathbb{E}[R_b].$$

EXPECTED RETURN ON A PORTFOLIO

Question

The portfolio return is

$$R_p = \omega R_s + (1 - \omega) R_b.$$

Consider ω fixed. What is the expected return?

Answer

Using E.3,

$$\mathbb{E}[R_p] = \omega \mathbb{E}[R_s] + (1 - \omega) \mathbb{E}[R_b].$$

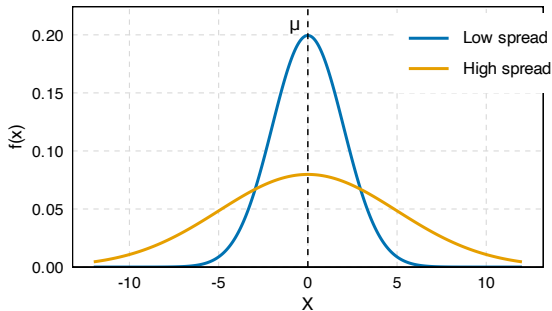
The expected return of a portfolio is just the **weighted average** of the pieces.

THE EXPECTED VALUE IS NOT ENOUGH

Consider you have two portfolios that give you the same expected return, but one distribution is more dispersed than the other.

Question

Which one would you take? Why?



THE EXPECTED VALUE IS NOT ENOUGH

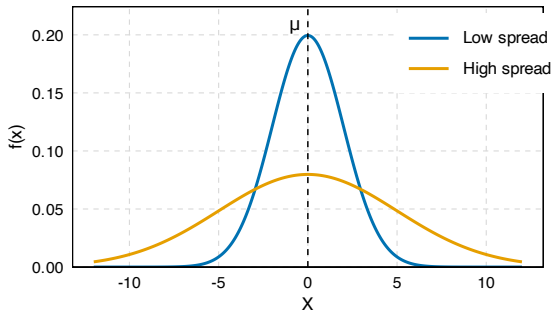
Consider you have two portfolios that give you the same expected return, but one distribution is more dispersed than the other.

Question

Which one would you take? Why?

Question

How to measure dispersion around the mean?



THE EXPECTED VALUE IS NOT ENOUGH

Consider you have two portfolios that give you the same expected return, but one distribution is more dispersed than the other.

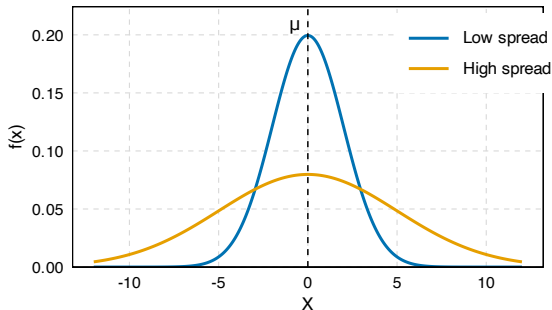
Question

Which one would you take? Why?

Question

How to measure dispersion around the mean?

Natural idea: measure the typical distance from X to its mean μ .



VARIANCE

Variance

$$\text{var}(X) = \mathbb{E}[(X - \mu)^2] = \sigma^2.$$

The expected **squared** distance from the mean.

VARIANCE

Variance

$$\text{var}(X) = \mathbb{E}[(X - \mu)^2] = \sigma^2.$$

The expected **squared** distance from the mean.

It is not the only candidate:

Mean absolute deviation

$$\mathbb{E}|X - \mu|$$

Median absolute deviation

$$\text{med}|X - \text{med}(X)|$$

Range

$$\max - \min$$

VARIANCE

Variance

$$\text{var}(X) = \mathbb{E}[(X - \mu)^2] = \sigma^2.$$

The expected **squared** distance from the mean.

It is not the only candidate:

Mean absolute deviation

$$\mathbb{E}|X - \mu|$$

Median absolute deviation

$$\text{med}|X - \text{med}(X)|$$

Range

$$\max - \min$$

Nice things about variance:

- **Smooth.** $\text{var}[X]$ is differentiable, which comes in really handy when deriving solutions to portfolio problems.

VARIANCE

Variance

$$\text{var}(X) = \mathbb{E}[(X - \mu)^2] = \sigma^2.$$

The expected **squared** distance from the mean.

It is not the only candidate:

Mean absolute deviation

$$\mathbb{E}|X - \mu|$$

Median absolute deviation

$$\text{med}|X - \text{med}(X)|$$

Range

$$\max - \min$$

Nice things about variance:

- **Smooth.** $\text{var}[X]$ is differentiable, which comes in really handy when deriving solutions to portfolio problems.
- **Punishes large deviations.** Points far from the mean count more towards the variance. Note that positive and negative deviations are treated equally.

VARIANCE

Variance

$$\text{var}(X) = \mathbb{E}[(X - \mu)^2] = \sigma^2.$$

The expected **squared** distance from the mean.

It is not the only candidate:

Mean absolute deviation

$$\mathbb{E}|X - \mu|$$

Median absolute deviation

$$\text{med}|X - \text{med}(X)|$$

Range

$$\max - \min$$

Nice things about variance:

- **Smooth.** $\text{var}[X]$ is differentiable, which comes in really handy when deriving solutions to portfolio problems.
- **Punishes large deviations.** Points far from the mean count more towards the variance. Note that positive and negative deviations are treated equally.

We usually denote the variance as a constant $\sigma^2 = \text{var}[X]$.

The variance is also referred to as the “second moment.”

STANDARD DEVIATION

Squaring created a problem. If X is a return in percent, then $\text{var}(X)$ is in **percent squared** — a unit that does not mean much.

STANDARD DEVIATION

Squaring created a problem. If X is a return in percent, then $\text{var}(X)$ is in **percent squared** — a unit that does not mean much.

Standard Deviation

$$\text{sd}(X) = \sqrt{\text{var}(X)} = \sigma.$$

STANDARD DEVIATION

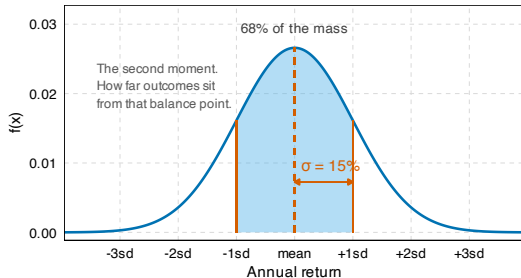
Squaring created a problem. If X is a return in percent, then $\text{var}(X)$ is in **percent squared** — a unit that does not mean much.

Standard Deviation

$$\text{sd}(X) = \sqrt{\text{var}(X)} = \sigma.$$

Taking the square root returns us to the **original units**.

A standard deviation of 15% per year is more meaningful because it carries the same unit as the mean. See graph.



PROPERTIES OF VARIANCE

VAR.1 $\text{var}(X) = 0$ exactly when X is a constant.

PROPERTIES OF VARIANCE

VAR.1 $\text{var}(X) = 0$ exactly when X is a constant.

VAR.2 $\text{var}(aX + b) = a^2 \text{var}(X)$.

SD.2 $\text{sd}(aX + b) = |a| \cdot \text{sd}(X)$.

Note what b does in VAR.2: **nothing**. A sideways shift moves the center, not the spread.

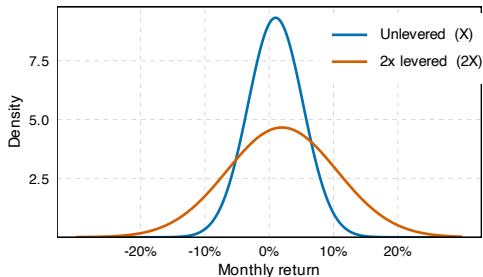
PROPERTIES OF VARIANCE

VAR.1 $\text{var}(X) = 0$ exactly when X is a constant.

VAR.2 $\text{var}(aX + b) = a^2 \text{var}(X)$.

SD.2 $\text{sd}(aX + b) = |a| \cdot \text{sd}(X)$.

Assume you take a leveraged position in the market, you borrow \$1 for every \$1 you invest out-of-pocket.



Leverage doubles your return (E.2) but SD.2 tells us that it also **doubles** the standard deviation.

PROPERTIES OF VARIANCE

VAR.1 $\text{var}(X) = 0$ exactly when X is a constant.

VAR.2 $\text{var}(aX + b) = a^2 \text{var}(X)$.

SD.2 $\text{sd}(aX + b) = |a| \cdot \text{sd}(X)$.

VAR.3

$\text{var}(aX + bY) = a^2 \text{var}(X) + b^2 \text{var}(Y) + 2ab \text{cov}(X, Y)$.

Notice the $\text{cov}(X, Y)$ term! It has a natural meaning in the context of portfolio returns. We'll discuss in next slides.

COVARIANCE

Covariance

$$\text{cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] .$$

One way to read this definition:

- covariance is **positive** when X and Y co-move around their means on average.
- covariance is **negative** when X and Y move in opposite directions around their means on average.

COVARIANCE

Covariance

$$\text{cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] .$$

Another way of writing covariance:

$$\text{cov}(X, Y) = \mathbb{E}[XY] - \mu_X \mu_Y .$$

One way to read this definition:

- covariance is **positive** when X and Y co-move around their means on average.
- covariance is **negative** when X and Y move in opposite directions around their means on average.

This equivalent formulation highlights that what really matters is $E[XY]$!

PROPERTIES OF COVARIANCE

COV.1 $\text{cov}(X, Y) = \text{cov}(Y, X)$.

PROPERTIES OF COVARIANCE

COV.1 $\text{cov}(X, Y) = \text{cov}(Y, X).$

COV.2 $\text{cov}(X, X) = \text{var}(X).$

PROPERTIES OF COVARIANCE

COV.1 $\text{cov}(X, Y) = \text{cov}(Y, X).$

COV.2 $\text{cov}(X, X) = \text{var}(X).$

COV.3 $\text{cov}(a_1X + b_1, a_2Y + b_2) = a_1a_2 \text{cov}(X, Y).$

PROPERTIES OF COVARIANCE

COV.1 $\text{cov}(X, Y) = \text{cov}(Y, X).$

COV.4 $|\text{cov}(X, Y)| \leq \text{sd}(X) \text{sd}(Y).$

COV.2 $\text{cov}(X, X) = \text{var}(X).$

COV.3 $\text{cov}(a_1X + b_1, a_2Y + b_2) = a_1a_2 \text{cov}(X, Y).$

PROPERTIES OF COVARIANCE

COV.1 $\text{cov}(X, Y) = \text{cov}(Y, X).$

COV.2 $\text{cov}(X, X) = \text{var}(X).$

COV.3 $\text{cov}(a_1X + b_1, a_2Y + b_2) = a_1a_2 \text{cov}(X, Y).$

COV.4 $|\text{cov}(X, Y)| \leq \text{sd}(X) \text{sd}(Y).$

COV.5 If X and Y are **independent**, then $\text{cov}(X, Y) = 0.$

COVARIANCE HAS A UNITS PROBLEM

Using monthly returns, 1941–2023, the *sample* covariance is

$$\widehat{\text{cov}}(R_s, R_b) = -0.0000265.$$

COVARIANCE HAS A UNITS PROBLEM

Using monthly returns, 1941–2023, the *sample* covariance is

$$\widehat{\text{cov}}(R_s, R_b) = -0.0000265.$$

Question

Is that a big number or a small one?

COVARIANCE HAS A UNITS PROBLEM

Using monthly returns, 1941–2023, the *sample* covariance is

$$\widehat{\text{cov}}(R_s, R_b) = -0.0000265.$$

Question

Is that a big number or a small one?

Answer

Impossible to say — quoted in percent, the same covariance reads -0.265 .

COVARIANCE HAS A UNITS PROBLEM

Using monthly returns, 1941–2023, the *sample* covariance is

$$\widehat{\text{cov}}(R_s, R_b) = -0.0000265.$$

Question

Is that a big number or a small one?

Answer

Impossible to say — quoted in percent, the same covariance reads -0.265 .

Question

What are the units?

COVARIANCE HAS A UNITS PROBLEM

Using monthly returns, 1941–2023, the *sample* covariance is

$$\widehat{\text{cov}}(R_s, R_b) = -0.0000265.$$

Question

Is that a big number or a small one?

Answer

Impossible to say — quoted in percent, the same covariance reads -0.265 .

Question

What are the units?

Answer

Return units **squared** — covariance carries the units of *both* variables.

COVARIANCE HAS A UNITS PROBLEM

Using monthly returns, 1941–2023, the *sample* covariance is

$$\widehat{\text{cov}}(R_s, R_b) = -0.0000265.$$

Question

Is that a big number or a small one?

Answer

Impossible to say — quoted in percent, the same covariance reads -0.265 .

Question

What are the units?

Answer

Return units **squared** — covariance carries the units of *both* variables.

Correlation

$$\text{corr}(X, Y) = \frac{\text{cov}(X, Y)}{\text{sd}(X) \text{sd}(Y)}$$

COVARIANCE HAS A UNITS PROBLEM

Using monthly returns, 1941–2023, the *sample* covariance is

$$\widehat{\text{cov}}(R_s, R_b) = -0.0000265.$$

Question

Is that a big number or a small one?

Answer

Impossible to say — quoted in percent, the same covariance reads -0.265 .

Question

What are the units?

Answer

Return units **squared** — covariance carries the units of *both* variables.

Correlation

$$\text{corr}(X, Y) = \frac{\text{cov}(X, Y)}{\text{sd}(X) \text{sd}(Y)}$$

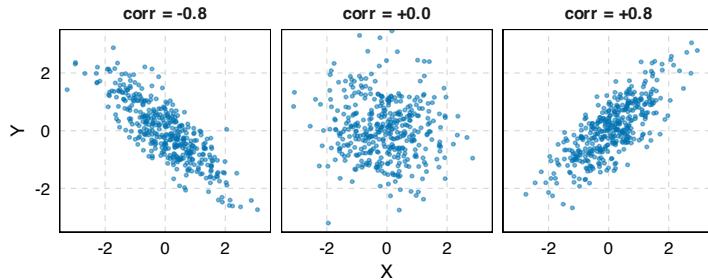
Nice properties of correlations:

- Dividing by both standard deviations cancels the units. Correlations are unit-less
- Correlations are always bounded:

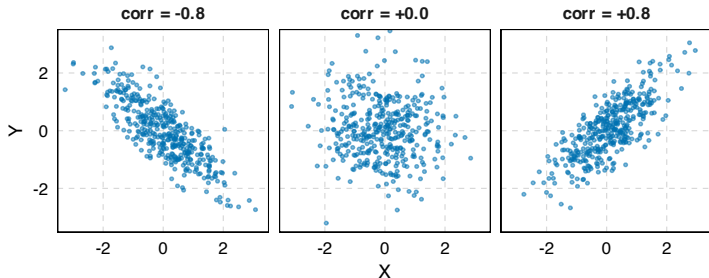
$$-1 \leq \text{corr}(X, Y) \leq 1$$

S&P Composite total return from Shiller; 10-year Treasury returns from the course data; 992 monthly observations.

WHAT CORRELATION LOOKS LIKE



WHAT CORRELATION LOOKS LIKE

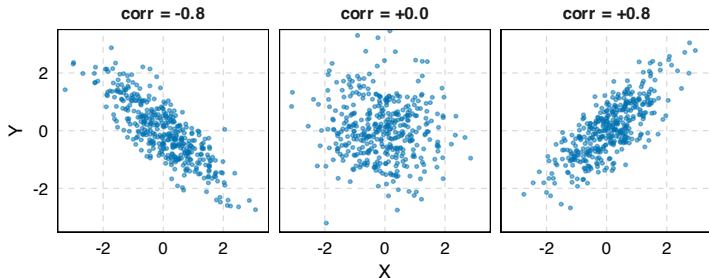


Near -1 : a tight downward line. One goes up, the other goes down.

Near 0 : a shapeless cloud. Knowing X tells you nothing linear about Y .

Near $+1$: a tight upward line. They move together.

WHAT CORRELATION LOOKS LIKE



Near -1 : a tight downward line. One goes up, the other goes down.

Near 0 : a shapeless cloud. Knowing X tells you nothing linear about Y .

Near $+1$: a tight upward line. They move together.

Correlation measures how close the cloud is to a **straight line**, and which way that line tilts.

VARIANCE OF PORTFOLIO RETURNS

Question

The portfolio return is

$$R_p = \omega R_s + (1 - \omega) R_b.$$

Consider ω fixed. What is the variance?

VARIANCE OF PORTFOLIO RETURNS

Question

The portfolio return is

$$R_p = \omega R_s + (1 - \omega) R_b.$$

Consider ω fixed. What is the variance?

Answer

Applying VAR.3,

$$\begin{aligned} \text{var}(R_p) &= \omega^2 \text{var}(R_s) + (1 - \omega)^2 \text{var}(R_b) \\ &\quad + 2\omega(1 - \omega) \text{cov}(R_s, R_b). \end{aligned}$$

If $\text{cov}(X, Y) < 0$, this means X and Y go in opposite directions, which implies that the total variance drops *below* the sum of the parts.

That is hedging

ANSWERING THE RISK–RETURN QUESTION

“Investors like high returns, but dislike high volatility” was translated as

$$\frac{\text{Return}}{\text{Risk}} = \frac{\text{Average return of } R_p}{\text{Volatility of } R_p}.$$

ANSWERING THE RISK-RETURN QUESTION

“Investors like high returns, but dislike high volatility” was translated as

$$\frac{\text{Return}}{\text{Risk}} = \frac{\text{Average return of } R_p}{\text{Volatility of } R_p}.$$

More formally, the risk-return profile of the portfolio is

$$\frac{\mathbb{E}[R_p]}{\text{sd}(R_p)}, \quad \text{sd}(R_p) = \sqrt{\text{var}(R_p)},$$

where

ANSWERING THE RISK-RETURN QUESTION

“Investors like high returns, but dislike high volatility” was translated as

$$\frac{\text{Return}}{\text{Risk}} = \frac{\text{Average return of } R_p}{\text{Volatility of } R_p}.$$

More formally, the risk-return profile of the portfolio is

$$\frac{\mathbb{E}[R_p]}{\text{sd}(R_p)}, \quad \text{sd}(R_p) = \sqrt{\text{var}(R_p)},$$

where

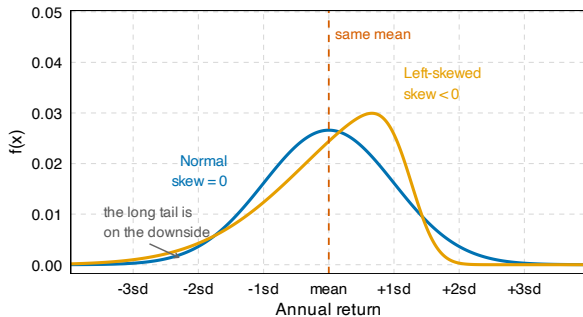
Numerator

$$\mathbb{E}[R_p] = \omega \mathbb{E}[R_s] + (1 - \omega) \mathbb{E}[R_b]$$

Denominator

$$\begin{aligned} \text{var}(R_p) = & \omega^2 \text{var}(R_s) + (1 - \omega)^2 \text{var}(R_b) \\ & + 2 \omega (1 - \omega) \text{cov}(R_s, R_b) \end{aligned}$$

THE THIRD MOMENT: WHICH SIDE IS HEAVIER

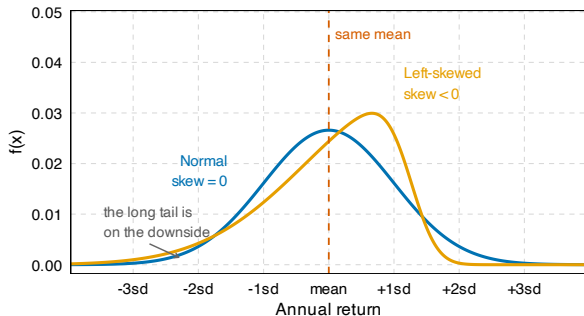


Skewness

$$\text{skew}(X) = \frac{\mathbb{E}[(X - \mu)^3]}{\sigma^3}$$

Symmetric distribution skew = 0

THE THIRD MOMENT: WHICH SIDE IS HEAVIER



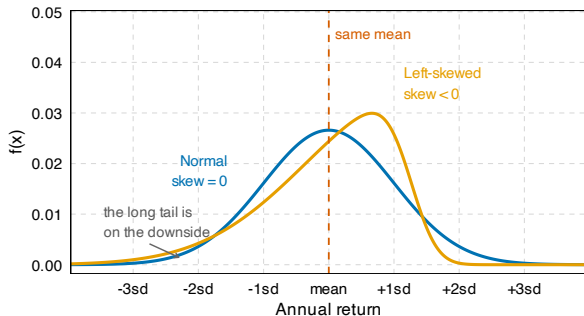
Skewness

$$\text{skew}(X) = \frac{\mathbb{E}[(X - \mu)^3]}{\sigma^3}$$

Symmetric distribution skew = 0

Negative skew: most months mildly good, a few catastrophically bad.

THE THIRD MOMENT: WHICH SIDE IS HEAVIER



Skewness

$$\text{skew}(X) = \frac{\mathbb{E}[(X - \mu)^3]}{\sigma^3}$$

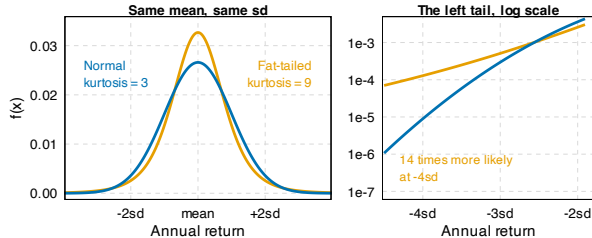
Symmetric distribution skew = 0

Negative skew: most months mildly good, a few catastrophically bad.

S&P monthly returns over 1941–2023:

Skew = **−0.89**

THE FOURTH MOMENT: HOW FAT THE TAILS ARE

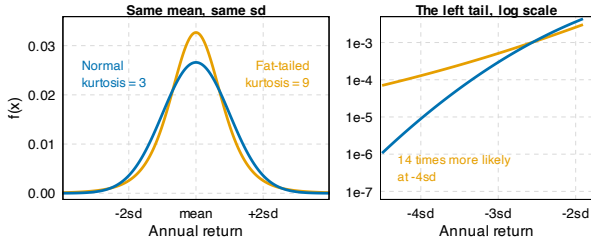


Kurtosis

$$\text{kurt}(X) = \frac{\mathbb{E}[(X - \mu)^4]}{\sigma^4}$$

Normal distribution kurtosis = 3.

THE FOURTH MOMENT: HOW FAT THE TAILS ARE



Kurtosis

$$\text{kurt}(X) = \frac{\mathbb{E}[(X - \mu)^4]}{\sigma^4}$$

Normal distribution kurtosis = 3.

S&P monthly returns over 1941–2023:

Kurtosis = **6.4**

KEY TAKEAWAYS

Expected value	$\mathbb{E}[X] = \sum_j x_j f(x_j)$ — the balance point of the distribution.
Mean $\neq$ expected value	$\bar{X}$ comes from data and is random; $\mathbb{E}[X]$ comes from the true distribution. $\bar{X} \rightarrow \mathbb{E}[X]$ (LLN).
Variance, sd	$\text{var}(X) = \mathbb{E}[(X - \mu)^2]$; σ returns it to the original units; captures volatility.
Covariance, correlation	cov captures comovement in variables. Correlation rescales cov into $[-1, 1]$.
The risk–return answer	$\mathbb{E}[R_p]$ is a weighted average. $\text{var}(R_p)$ is <i>not</i> ; cov is important in reducing volatility.
Higher moments	Skewness (3rd) and kurtosis (4th): which tail is long, how much lives far out. Normal: 0 and 3

REFERENCES

Shiller, Robert J. 2024. “U.S. Stock Markets 1871–Present and CAPE Ratio.” Online Data, <https://shillerdata.com>.